

Flow dynamic in a packed bed filled with Ni-Al₂O₃ porous catalyst: experimental and numerical approach

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Summary

Numerical simulations and experiments have been carried out to explore the effect of particle shapes on the pressure drop and the loss factor in a packed-fixed bed filled with a porous catalyst. The packed bed is presented by the Ni-Al₂O₃ catalyst of the different shapes. The commercial program ANSYS Fluent is used for the analysis; more than 20 mln cells are used for CFD modeling. The catalyst particles were set as a porous medium with the viscous resistance coefficients and the inertial resistance coefficients. The comparison of the pressure drops between the experimental and simulation results show a good correlation with the divergence of results less than 8%. To determine the effect of the porosity properties of the medium on the numerical results, two cases of CFD modeling were realized (with taking into account the porous medium properties and without it). The discrepancy between results increases with an increasing gas flow rate.

KEYWORDS:

catalyst, packed bed, CFD modeling, pressure drop, particle shape

INTRODUCTION

Steam methane reforming is an important process in various industries. A wide application of this process is found in the chemical industry for the production of hydrogen and synthesis gas^{1,2,3,4}. The majority of hydrogen (approximately 48%) is produced by steam reforming or partial oxidation of methane^{5,6}. In addition, in recent years, a wide interest among engineers and scientists of different countries has produced a way to increase the energy efficiency of fuel-consuming equipment as thermochemical waste-heat recuperation by steam methane reforming^{7,8,9}.

Usually, the steam methane reforming process is realized in methane reformers that are filled randomly with the catalyst of various shapes: spheres, cylinders, the Raschig rings, multi-hole cylinders, etc. The chemical reactions of overall steam methane

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as
doi: 10.1002/aic.16558

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Received: Sep 30, 2018; Revised: Jan 20, 2019; Accepted: Jan 28, 2019

reforming (SMR) in the methane reformer are well studied, both theoretically^{10,11,12,13} and experimentally^{14,15,16,17}. It is known that different types of catalysts are applied to improve the efficiency of this chemical process. The nickel (Ni) based catalysts are widely used in large-scale hydrogen production^{14,18,19}. Moreover, the noble metals such as ruthenium (Ru)²⁰, rhodium (Rh)²¹, palladium (Pd)⁵, iridium (Ir)¹⁰, and platinum (Pt)²² are also used as active elements in the catalysts for steam methane reforming.

The phenomena occurring in the tubes of a steam methane reformer are the endothermic reactions and their effects on composition, density, velocity, temperature, pressure drop effects, etc. The main goal of this article is to detail study the flow dynamics in a packed bed of the reformer.

Previous experimental investigations of the pressure drop are performed for various conditions: the shape of the catalyst, the porosity, the filling method, etc^{23,24}. The experimental setup for the pressure drop investigation through a packed bed is suitably simple and consists mainly of two manometers, a flow meter or a Pitot tube, a thermometer, etc²⁵. For the numerical investigations there are two major methods to determine the pressure drop of the reformer packed beds. In the first case, the packed bed is presented as a pseudohomogeneous media, where modified Navier-Stokes equations are used in conjunction with the Ergun pressure drop correlation to account for the fluid-solid interaction^{26,27,28}. In the second, the packed bed is modeled based on using the real geometries of the packed bed and catalyst shape. This approach provides a more detailed description of the gas flow in the packed bed. In the second approach, the Navier-Stokes equations are used to the fluid domain between the catalyst particles^{25,29,30}.

In recent years, a numerical investigation of the different physical and chemical processes has found wide use among researchers. CFD-modeling allows using a realistic packing structure for the packed bed and simulates flow dynamics between catalyst particles. The main advantages of CFD-modeling of flow dynamics in the packed bed are the low cost of the investigation in comparison with the experiments, as well as an evident visualization of the results. The different commercial and open-source codes are applied for CFD simulation of flow in the packed bed, for example, ANSYS Fluent^{31,32,33,34,35}, Comsol Multiphysics³⁶, Autodesk Simulation CFD³⁷, OpenFOAM (open source), etc.

Dixon and Nijemeisland reported about CFD modeling as a design tool for a fixed-bed reactor²⁹. The authors concluded that CFD-modeling of a gas flow through the reformer with the packed bed can provide highly detailed and reliable information about the flow fields. They simulated the gas flow for a moderate simple geometry with a low number of the cells in a mesh and the catalyst particles. The exponential growth of the computer power provided a supplementary opportunity for obtaining more detailed and accurate information in the numerical results. On Ref.²⁹, the catalyst packed bed is filled by the spherical particles. Additionally, most of the studies of flow dynamics in the packed bed consider only simple shapes of the catalyst particles: spherical^{29,30,24} and cylindrical²⁴.

Bai and co-workers presented the numerical simulation results of the flow fields distributions and the pressure drop in a fixed bed reactor with randomly packed catalyst particles²⁴. They reported the numerical results where the discrete element method

(DEM) and computational fluid dynamics (CFD) are coupled to model a fixed bed reactor with low tube-to-particle diameter ratios ($D/d < 4$). The authors considered spherical and cylindrical shapes of catalysts and compared the obtained numerical results with experimental data. They showed a good correlation between the CFD-modeling and experimental results, but they concluded that with an increasing gas flow rate in the packed bed the divergence between results also increased. The packed bed was presented as a solid body.

In most of the research mentioned above, the catalyst particles are considered as solid bodies but it is known that the catalyst for steam methane reforming is a porous medium. Wehinger and co-authors³⁸ reported that the pore processes have a detectable effect on flow dynamics in the packed bed. They numerically investigated flow dynamics in the porous medium of the catalyst bed via the CFD software STAR-CCM+ for a wide range of Reynolds number from 10 to 2000. Moreover, CFD modeling of gas flow in the porous medium was presented by Gao and Zhu³⁹ to simulate the flow behaviour and catalytic coupling reaction of carbon monoxide (CO) – diethyl oxalate (DEO). They analyzed a two-dimensional computational domain. Karthik and co-worker²³ set the catalyst bed as a porous medium but they worked with a simple geometry of a packed bed with the number of catalyst particles being very low.

The aim of this study is to develop a computational fluid dynamics (CFD) model of flow dynamics in the fixed packed bed for the real geometry of the packed bed taking into account porosity of the catalysts particles. To achieve this aim, meshes with more than 20,000,000 cells are used. To verify the numerical results, the experimental determination of the pressure drop in the reformer is performed for the various shapes of catalyst particles. In this study four shapes of the catalyst particles are considered: cylinder, a cylinder with internal holes, a Raschig ring, and a sphere with internal holes. The experimental and numerical study are performed for the different initial conditions. The commercial ANSYS Fluent CFD code is used for the numerical simulation.

With the modern computational tools and high power computing technology available, the computational fluid dynamics simulations to predict the pressure drop of the catalyst bed of the steam methane reformer are feasible. Whether they will partly or fully replace experimental investigations will largely depend on the reliability of the turbulence and porous medium models that are used, and the comprehensive study of them. Therefore, both numerical and experimental investigation of the flow dynamics process in a fixed-bed reformer is a crucial task.

EXPERIMENTS

Experimental setup

The experimental setup shown in Fig. 1 is used to study fluid dynamics and the pressure drop of the packed bed. The experimental setup consists of the following elements: 1 - an air heater ; 2 - the manometer tubes; 3 - a ceramic cylindrical tube filled with

catalysts; 4 - an instrument board presenting the pressure and flow temperature; 5 - a transparent tube with catalysts (to show packing structure); 6 - a fan/ventilator

To visualize the packing structure of a catalyst bed, the catalyst particles are placed to a transparent acrylic tube (5). Then packed bed is placed in the experimental setup. The gas flow is controlled by a fan rotor speed and measured by a Veterok GRM-12 flow-meter. The electronic manometers are used to measure pressure before and after the packed bed. The measuring positions of the electronic manometers were more than 3 times higher than the tube diameter to ensure stable pressure data. SMR is usually performed at temperatures up to 700-800 °C. Therefore, air is heated up to 700 °C by air heater.

Each structure of the packed bed is tested under 10-15 different flow rates. For each flow rate, the measurements are repeated at least 3-5 times to quantify experimental precision. The pressure drop is determined as the average of the repetitions. All experiments are performed at air temperature about 700 °C.

Catalyst shapes and packed bed

The packed beds with a structured packing of catalyst particles with the different shapes are presented in Fig.2. There are four particle shapes: A – a simple cylinder; B – a Raschig ring (cylinder with one hole on axial line); C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes.

The catalyst particles are placed in the tube with a diameter of 100 mm. The catalysts layer depth is 600 mm. The performances of catalysts are shown in Table 2. For experiment the Ni-Al₂O₃ catalyst is used with the following chemical composition: NiO = 14.5%; SiO₂ = 0.2%; support CaO-MgO-La₂O₃-Al₂O₃. Other properties are presented below: bulk density is 680 kg/m³; surface area is 4.5 m²/g_{cat}; average porosity of catalyst particles is about 41%.

CFD MODELING

Geometry and mesh

In the present paper, the family of particle shapes is considered as it is shown in Fig.2.

The computational domain has the maximum similarity to the experimental package of catalysts in the reformer. For this numerical investigation, it is assumed that the size of all catalyst particles is the same. In reality, the diameter of the sphere can be different by 0.2-0.4 mm. The computational domain of the packed bed is presented in Fig.2. The packed bed is 600 mm in length, the diameter is 100 mm. All the cases have the same outer diameter and length. To obtain a developed flow for experiments, when a laminar flow takes place, free space is added before the packed layer. The length of this free space is 250 mm.

The mesh structures for all investigated computational domains are presented in Fig.4. The edge size of the elementary cells is the same for all computational domains. The meshes for all cases are generated with the same mesh settings. The inflation

layers around the catalyst were added in the fluid domain to get a more perfect grid. The inflation layer is also added near the tube wall in the fluid domain. The total thickness of the inflation layer is adjusted to get $20 < y^+ < 250$, a requirement for $k-\epsilon$ turbulence model. The mesh independence study is not performed. The maximum number of elementary cells (about 23 million for our university ANSYS license) are used.

Governing equations

Single phase, turbulent, steady flow is modeled by solving the Reynolds-averaged Navier-Stokes equations. The governing equations are solved in ANSYS Fluent. The assumptions are taken for CFD-modeling: the steady-state conditions; no flux of energy due to a mass concentration gradient (no Dufour effects)⁴⁰; work by viscous forces and by pressure is not done.

The equation for the conservation of mass, or continuity equation, for steady-state conditions, can be written as follows:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (1)$$

For the fluid domain the conservation of momentum is described by the following equations⁴¹:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\bar{\tau}}) + \rho \vec{g} + \vec{F} \quad (2)$$

where p is the static pressure; $\rho \vec{g}$ is the gravitational body force.

The stress tensor $\bar{\bar{\tau}}$ is given by the following equation:

$$\bar{\bar{\tau}} = \mu \left[(\nabla \vec{v} + \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (3)$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The porous media domain is simulated by the addition of source term for the equation (2) that can be expressed as:

$$S_i = - \left(\sum_{j=1}^3 D_{ij} \mu v_j + \sum_{j=1}^3 C_{ij} \frac{1}{2} \rho |v| v_j \right) \quad (4)$$

The momentum source term S_i for every dimension (x, y, z) consists of two parts. The first term of the equation (4) is a viscous loss term. The second term of the equation (4) is an inertial loss term. This momentum sink contributes to the pressure gradient in the porous cell, creating the pressure drop that is proportional to the fluid velocity (or velocity squared) in the cell.

The presented model is solving the standard conservation equations for turbulence quantities in the porous medium. In this default approach, turbulence in the medium is treated as though the solid medium has no effect on the turbulence generation or dissipation rates. This assumption may be reasonable if the medium's permeability is quite large and the geometric scale of the medium does not interact with the scale of the turbulent eddies.

Typically, steam methane reforming is taking place at a wide range of Reynolds numbers. For the numerical model, the RNG k - ϵ model is chosen^{41,42,43}. The Scalable Wall Function is used for RNG k - ϵ model because there are large numbers of contacting surfaces in the computational domain. The mathematical description of RNG k - ϵ model can be expressed as follows:

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \epsilon - Y_M + S_k \quad (5)$$

and

$$\frac{\partial}{\partial x_i}(\rho \epsilon v_i) = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon + S_\epsilon \quad (6)$$

In equations for k - ϵ turbulence model, G_k shows the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. The quantities α_k and α_ϵ are the inverse effective Prandtl numbers for k and ϵ , respectively. S_k and S_ϵ are user-defined source terms.

The additional term R_ϵ in the ϵ equation can be determined as follows:

$$R_\epsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0) \epsilon^2}{1 + \beta \eta^3} \frac{\epsilon^2}{k} \quad (7)$$

where $\eta \equiv Sk/\epsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

The model constants $C_{1\epsilon}$ and $C_{2\epsilon}$ in k - ϵ turbulence model used by default setup for Fluent solver of ANSYS⁴⁴:

$$C_{1\epsilon} = 1.42; \quad C_{2\epsilon} = 1.68 \quad (8)$$

The performances of the porous medium are determined by the viscous resistance coefficients and the inertial resistance coefficients for each direction. These coefficients, respectively, can be determined as follows:

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon^3}{(1 - \epsilon)^2}; \quad (9)$$

$$C_2 = \frac{1.75 \rho (1 - \epsilon)}{D_p} \frac{1}{\epsilon^3}, \quad (10)$$

where D_p – the length of the fixed bed; ϵ – the void fraction. Here the void fraction (ϵ) is taken into account the interparticle spaces and the pellet porosity.

The schematic algorithm of CFD modeling is presented in Fig.5.

RESULTS AND DISCUSSION

Verification

The experimental and numerical investigation was performed at the same operating conditions. The CFD model was realized via ANSYS Fluent on Xeon E5-2670 Sandy Bridge-EP server with double 20-core 2.6 GHz processors with 32 GB RAM. Convergence for each numerical experiment took approximately 3000 iterations for which the computation time was on the order of 20-23 hours. The Reynolds number during both experimental and numerical investigations ranged from about 500 to about 9000. The change in the Reynolds number is done by changing the velocity.

All the experimental data are denoted by the hollow points. The results of the numerical simulation are represented by the solid lines. To verify the results obtained with the help of CFD model, they were compared with experimental data for the different packed beds. Fig.6 shows the comparison of numerical and experimental data for the different packed beds with 600 mm in length. The value of the pressure drop for the numerical results was obtained by Vortex Average function in the Fluent solver.

As can be seen from Fig.6, the differences between the experimental and numerical results are observed for all investigated cases. However, at the velocity range above 0.5 m/s, which corresponds to the real operation parameters of the steam methane reformers, the discrepancies between the results are reduced. The average deviations between the numerical and experimental results in the velocity range above 1 m/s are 6.12% for the cylinder without holes, 2.15% for the Raschig ring, 4.75% for the cylinder with 7 holes, 4.34% for the sphere with 7 holes. Therefore, it can be concluded that the results have a good correlation and the mathematical model is reliable to study the flow dynamics in the packed bed.

Pressure and velocity contours

Fig.7 shows the typical CFD simulation results for the packed bed filled with the cylindrical catalysts, including the velocity contours on the catalyst particles (Fig.7A), the velocity contours into the catalyst particles (Fig.7B) and the pressure contours on particles (Fig.7C). Each point on the pressure profile plot corresponds to one spatial position in the packing. Fig.7 depicts that the flow of gas passes not only between the catalyst particles but inside them.

The velocity contours in the axial and radial cross-section of the packed bed and are shown in Fig.8. The velocity magnitude is significantly lower into the particle shapes compared to the velocity magnitude into free space between the cylindrical particles.

Loss factor

Fig.9 shows the dependence of the pressure losses per meter of the packed bed for different Reynolds numbers. From Fig.9, it can be seen that the maximum pressure loss per 1 meter of the packed bed takes place for a cylinder without holes (case A in Table1), while a minimum pressure loss occurs for a cylinder with 7 holes (case C in Table1). At the same time, a characteristic

quadratic dependence $\Delta P = f(Re^2)$ is observed for all four curves: $\Delta P = f(Re^{1.92})$ for A case; $\Delta P = f(Re^{1.94})$ for B case; $\Delta P = f(Re^{1.93})$ for C case; $\Delta P = f(Re^{1.94})$ for D case.

From the point of view of the classical representation of fluid dynamics, the pressure losses in the catalyst bed can be considered as the local hydraulic losses, which are caused by changes in the shape and size of the channel, which is deforming the flow. As shown by the experimental and numerical results (Fig.6 and Fig.9), it can be assumed that the pressure loss through the catalyst bed is proportional to the square of the velocity. For the engineering calculations, it is convenient to characterize the fluid resistance of a catalyst bed with a dimensionless quantity K_l , which is called the local resistance coefficient or loss factor.

The local resistance coefficient can be determined experimentally:

$$K_l = \frac{2\Delta P}{\rho v^2}, \quad (11)$$

where ρ – the density of fluid; v – the velocity of fluid.

To determine the local resistance coefficient, the effect of length of the packed bed on the pressure drop is investigated. For this purpose, the different lengths of the packed bed are considered. The computational domains (Fig.3) were changed, but the mesh structures were the same as in previous cases. Fig.10 shows the variations of the pressure drop as a function of the packed bed length for $Re=6000$.

As it can be seen from Fig.10, the dependence of the pressure loss for the different lengths of the catalyst bed is approximately linear. This fact allows concluding that the value of the local resistance coefficient for the catalyst bed is determined by the amount of catalyst in the packed bed and the shape of the catalyst particle.

Based on the results which are presented in Fig. 10, and also on the basis of the expression (11), the values of the local resistance coefficients were determined for various cases (Table1). The variations of the local resistance coefficient for 1 meter of the packed bed versus the average velocity for the different packed beds are shown in Fig.11.

As seen in Fig.11, the loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s. Usually, when flowing around solid non-porous bodies, the coefficient K_l depends weakly on the velocity. This effect of velocity on the coefficient K_l can be explained by the fact that with increasing velocity, the amount of gas that passes through the pores in a catalyst increases. At the same time, when the gas velocity is low, the fluid energy is not enough to percolate through the porous structure. The next paragraph presents the results of the investigation the velocity distribution in porous media.

Porous media

There are scientific articles dealing with the study of flow dynamics in a packed bed that treat catalyst particles as a solid body²⁴. In this paragraph, the effect of taking into account the porous medium properties on the numerical results is reported. For this

purpose, the computational domains that presented above were considered. The two types of the packed beds is considered: A – cylinder without holes; D – sphere with 7 holes.

Fig.12 show the discrepancy between the results obtained with taking into account the porous medium properties (solid line) and without it (dashed line). As it can be seen from Fig.12, the discrepancy between the results is increasing with an increase of the velocity magnitude, while the results the low velocity are almost similar. Therefore, for saving calculation time at low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

CONCLUSION

The numerical simulation and experimental investigation of fluid dynamics in the packed bed of $\text{Ni-Al}_2\text{O}_3$ catalysts with the different shapes are performed in this paper. The CFD model is realized via ANSYS Fluent, the Academic Research version which allows using large numbers of catalysts particles without any limitation. Both numerical and experimental studies were performed for the large L/d ratio of the packed bed for a wide range of Reynolds number. The algorithm of the fluid flow simulation, taking into account the porous medium properties of catalyst particles, is depicted. Four types of catalyst shapes are considered. A good correlation between the numerical and experimental results is achieved for all experiments with an average error of less than 8%.

A characteristic quadratic dependence $\Delta P = f(v^2)$ is observed for all catalyst types. The effect of the packed bed depth on the pressure drop is investigated. The dependence of the pressure loss for the different depths of the catalyst bed is approximately the same. This fact allowed to conclude that the local resistance coefficient for the catalyst bed is constant for all investigated L/d ratios. The loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s.

To determine the effect of porosity properties of the medium on numerical results, two cases of CFD modeling were realized. First, taking into account the porous medium properties and second, without it. The discrepancy between the results increases with an increase in the velocity, while for the low velocity the results are almost similar. As a recommendation for the engineering solutions, it can be concluded that for saving calculation time at the low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

ACKNOWLEDGMENTS

The author gratefully acknowledges the NIAP-CATALYZATOR LLC company for providing the hardware and software support for this research. This work was carried out with financial support from the grants of the President of the Russian Federation (MK-447.2019.8).

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FIGURE 1 Photograph of a laboratory experimental setup for investigation of fluid dynamics and pressure drop

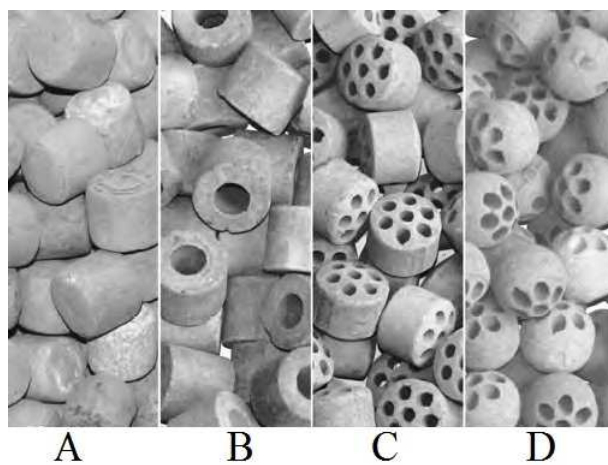


FIGURE 2 Photograph of the packed beds filled by the catalysts with different shapes: A – a cylindrical; B – a Raschig ring; C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes

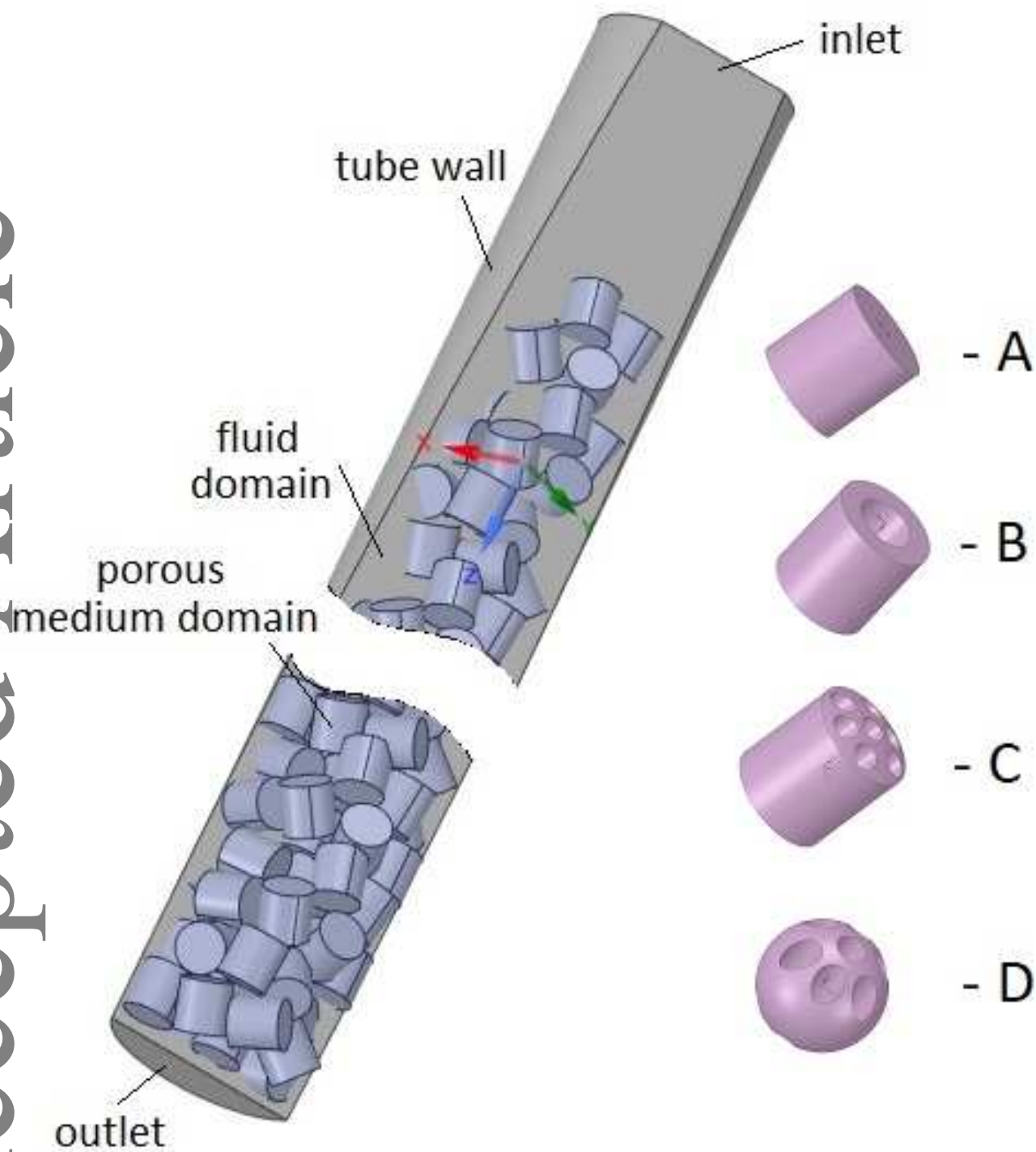


FIGURE 3 A computational domain of a packed bed that filled by the catalyst with cylindrical shape (A).

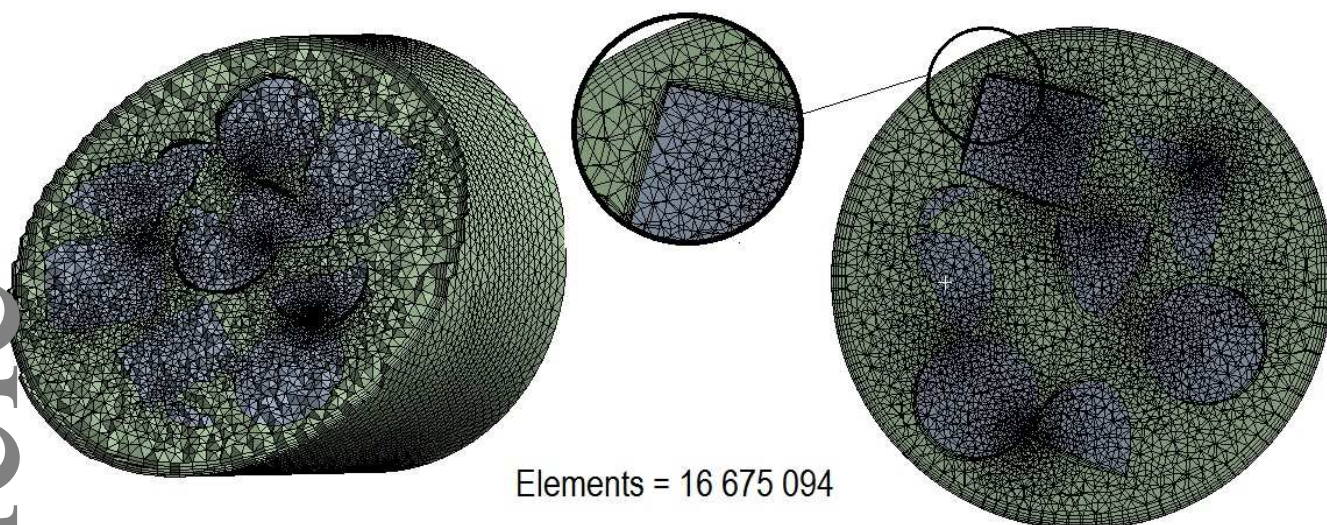


FIGURE 4 The mesh structure: a) catalyst porous media domain; b) fluid domain

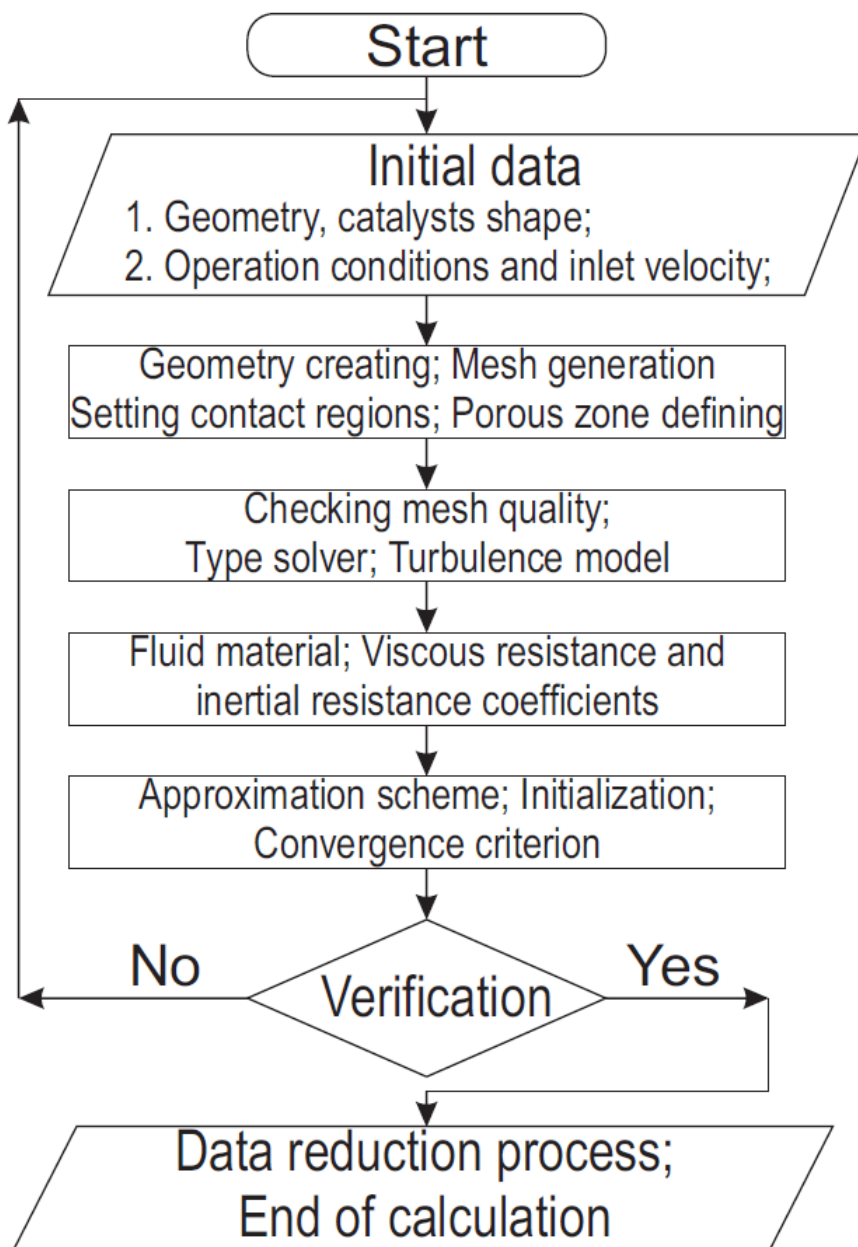


FIGURE 5 The algorithm of a fluid flow simulation in the packed bed.

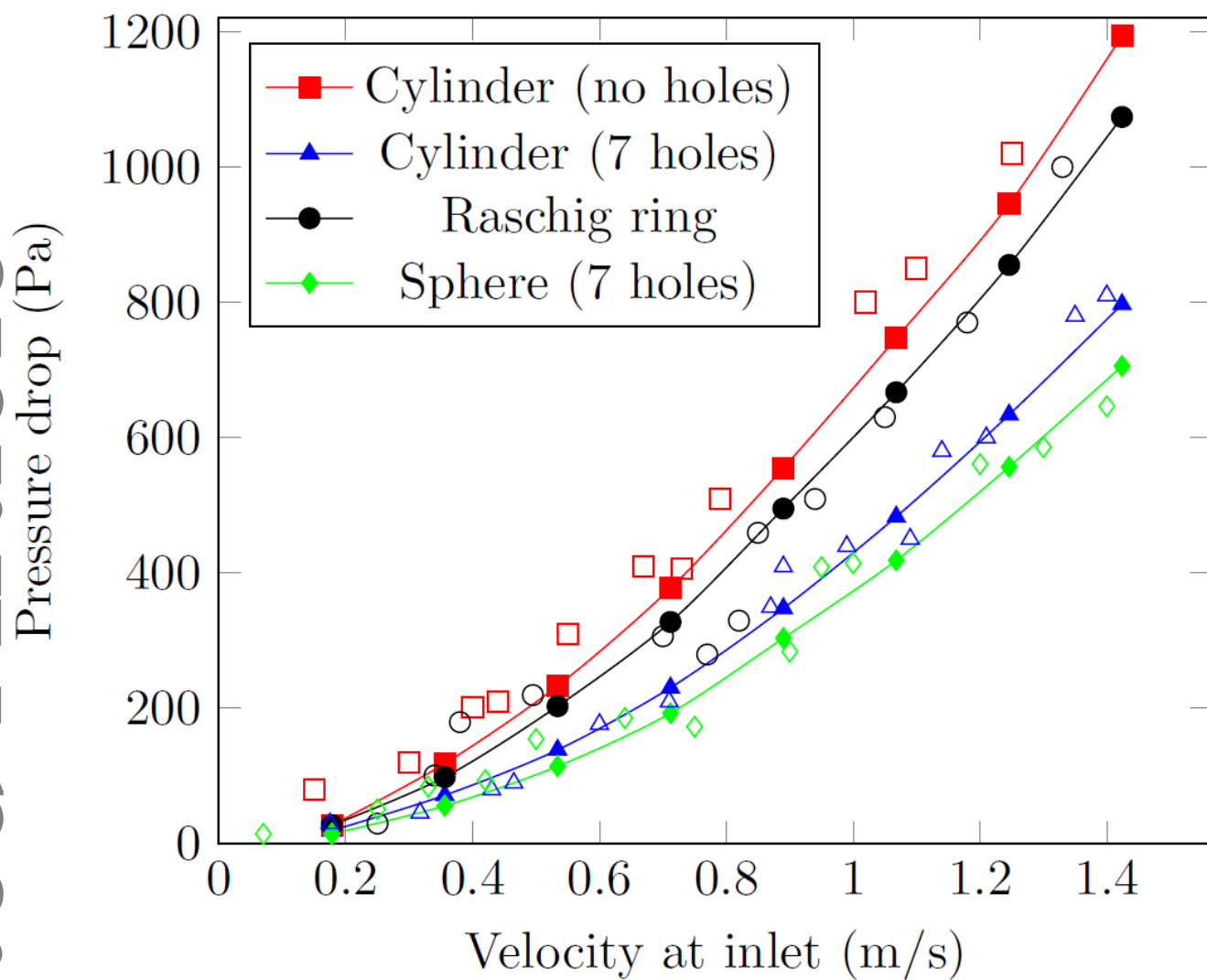


FIGURE 6 The comparison of numerical and experimental data: the hollow points – experiment; the solid points and lines – CFD modeling.

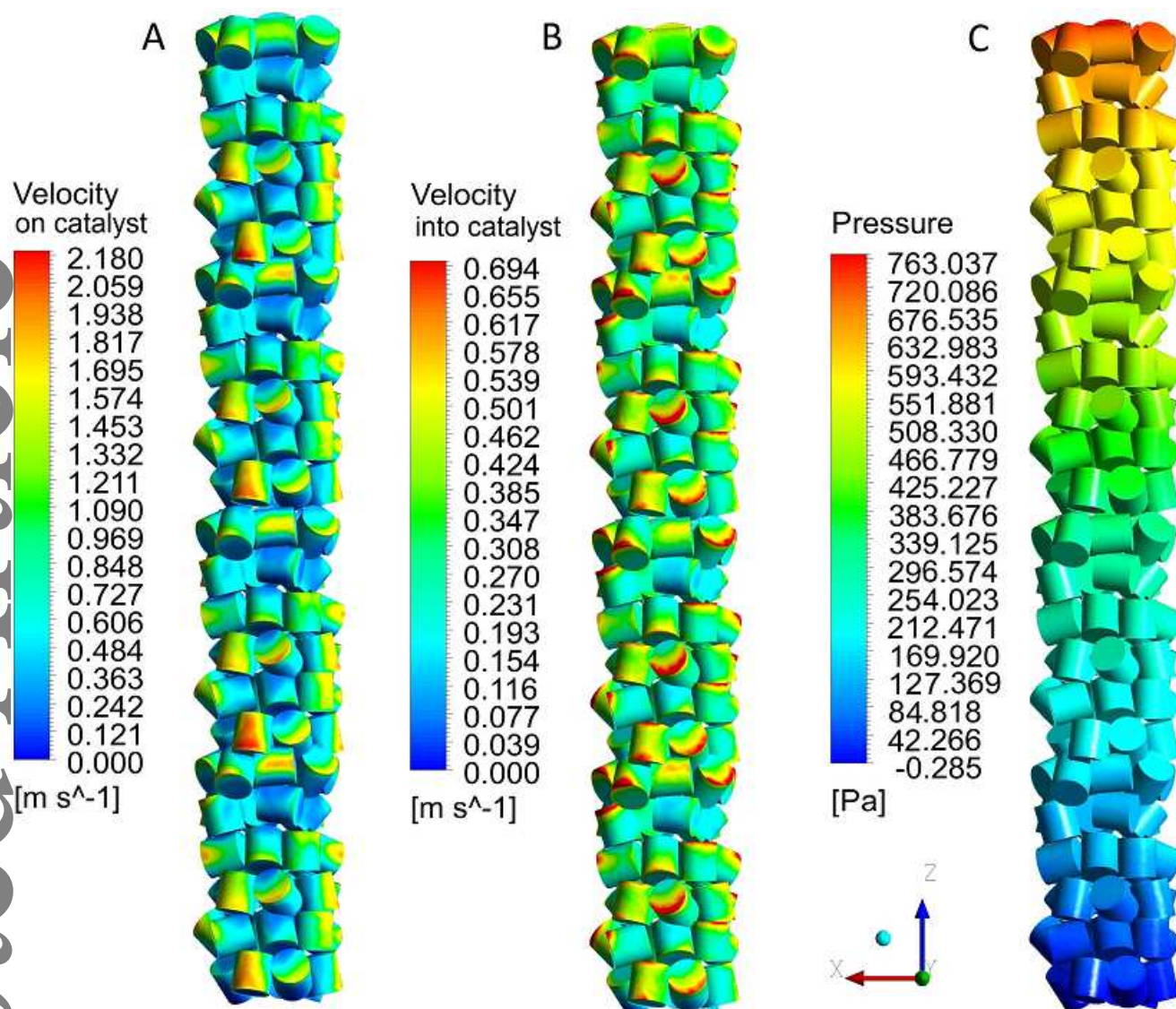


FIGURE 7 The velocity and pressure contours.

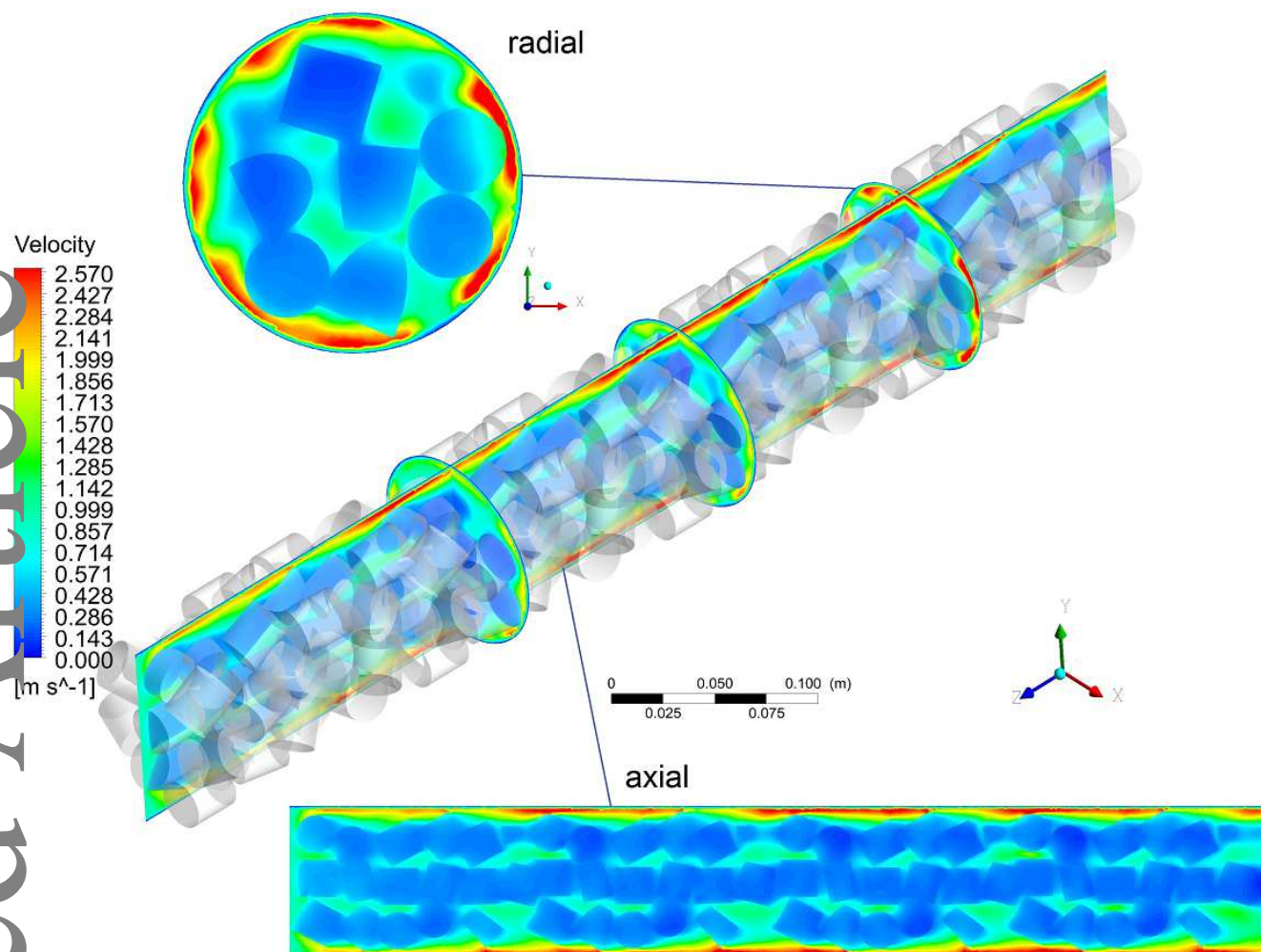


FIGURE 8 Simulated velocity magnitude distribution for velocity $u=0.7$ m/s.

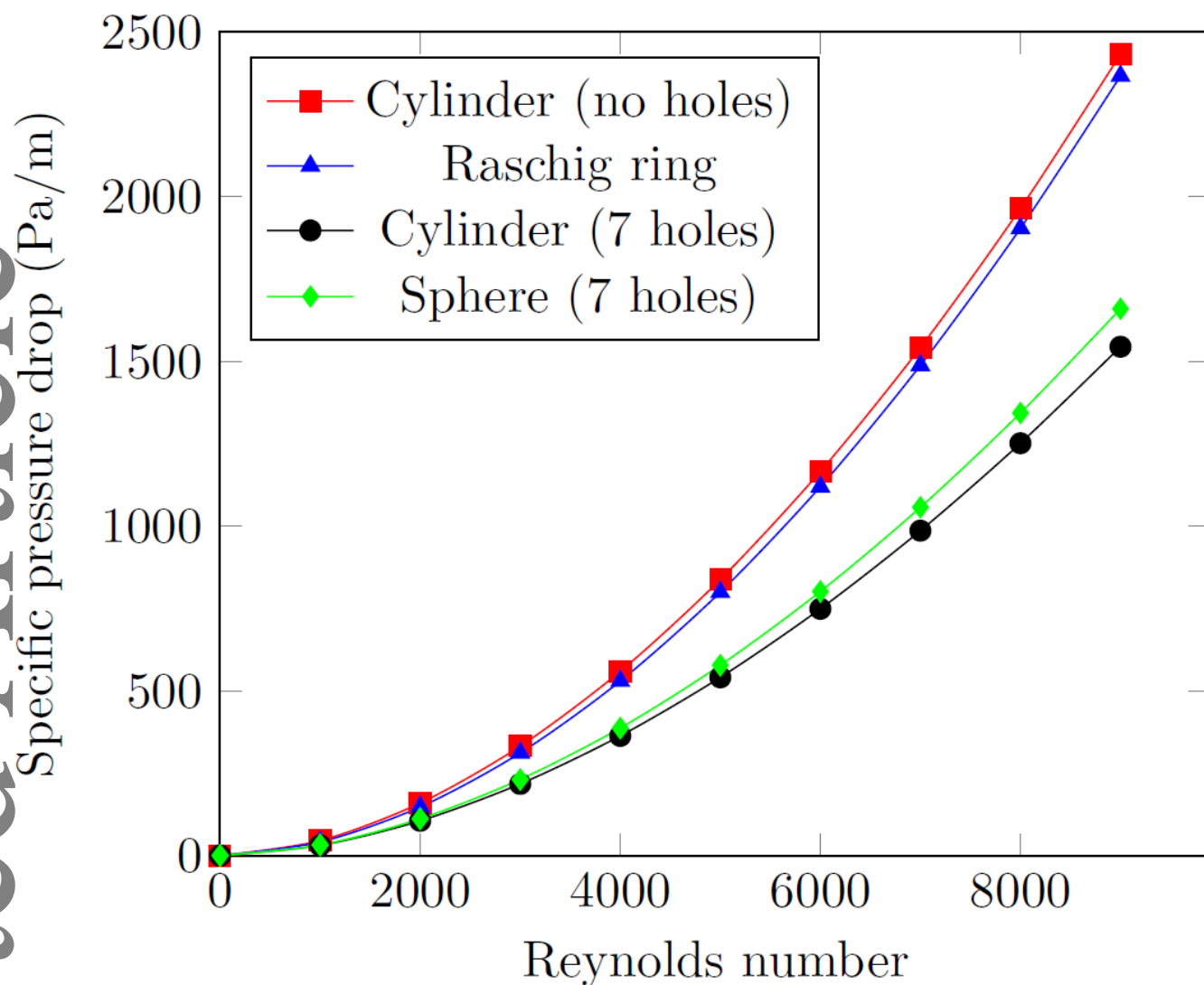


FIGURE 9 The variations of the pressure drop versus Reynolds number for the various packed beds.

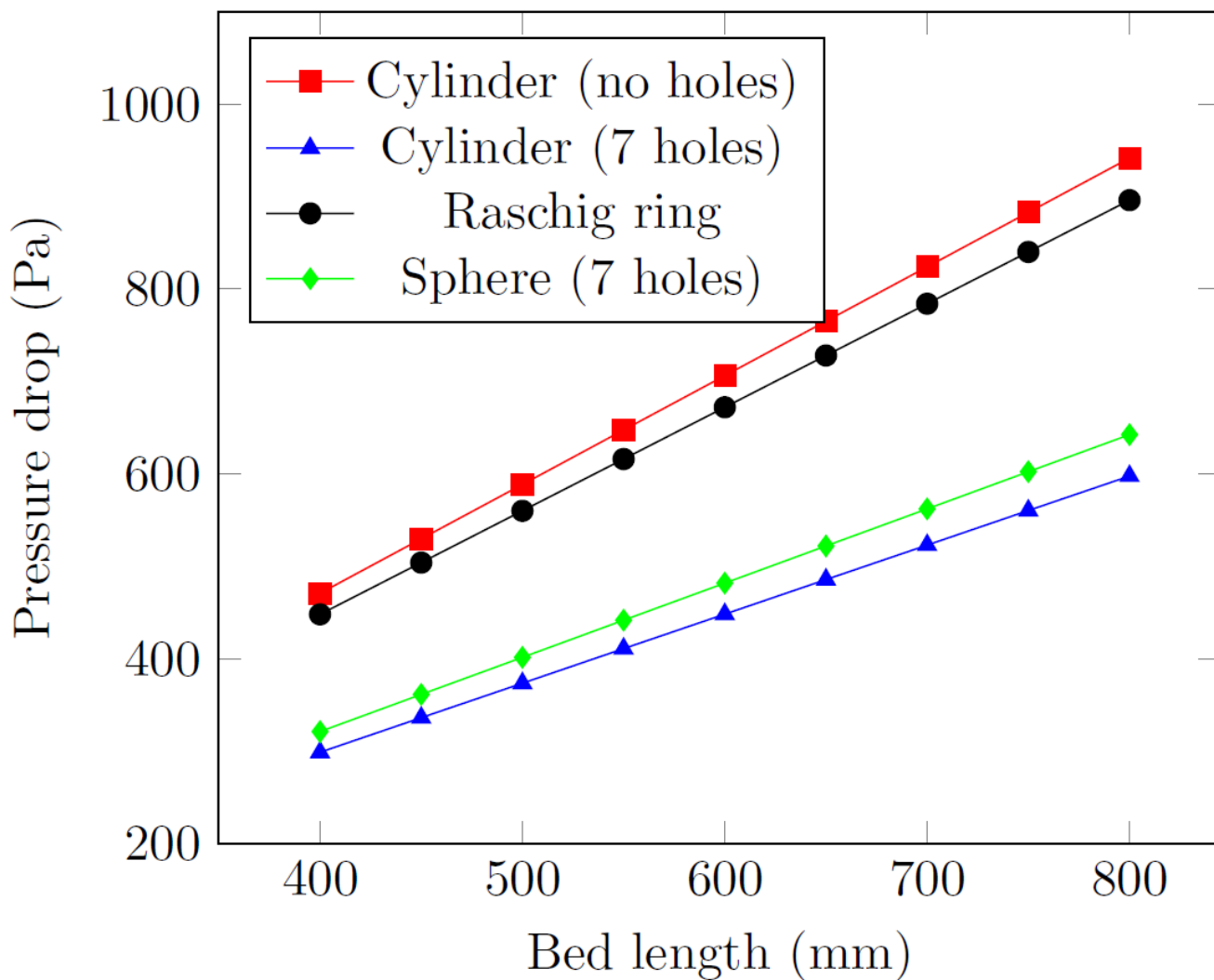


FIGURE 10 The variations of the pressure drop for the different packed beds length for $Re=6000$.

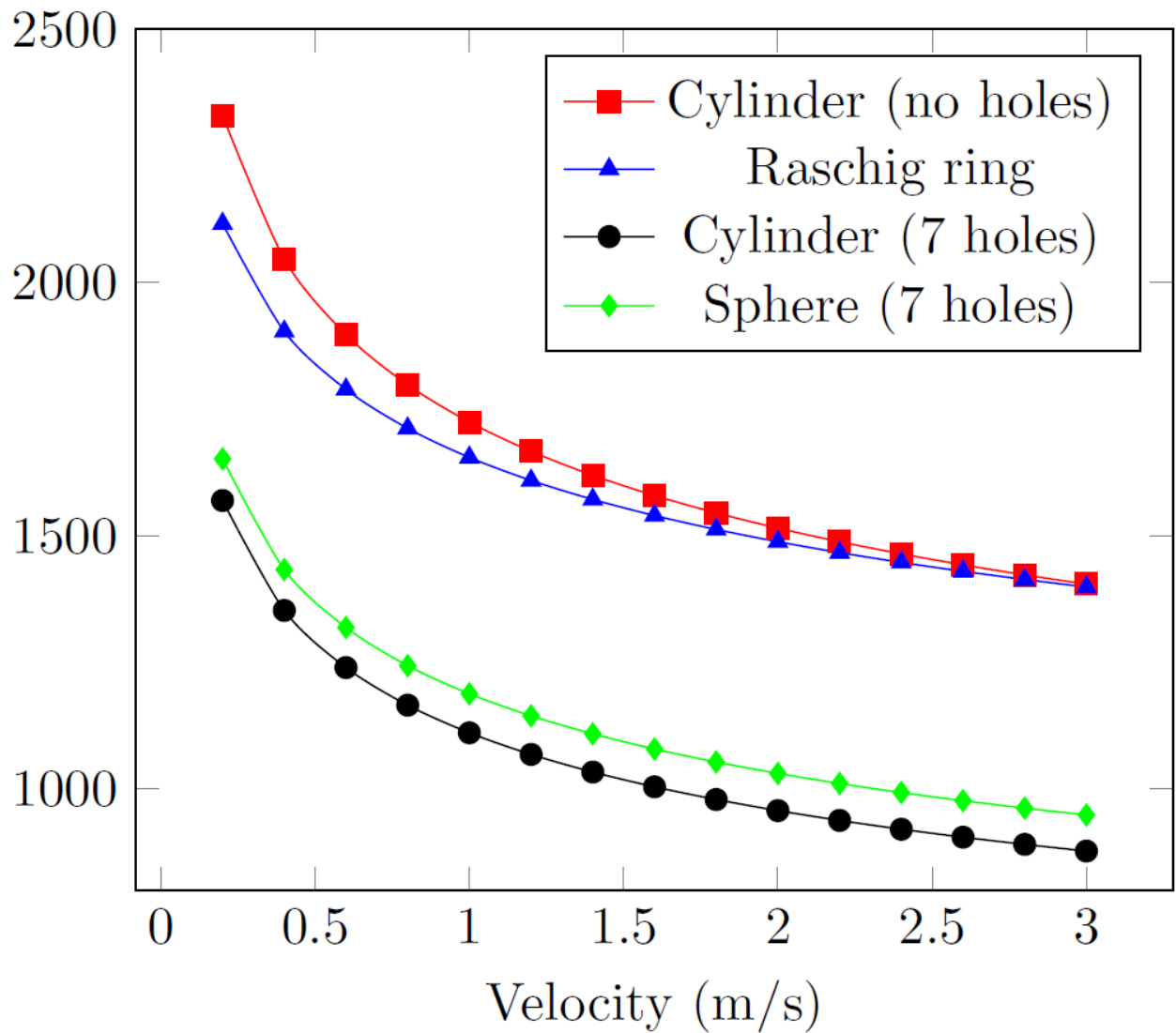


FIGURE 11 The variations of the local resistance coefficients versus the velocity for the different packed beds for $Re=6000$.

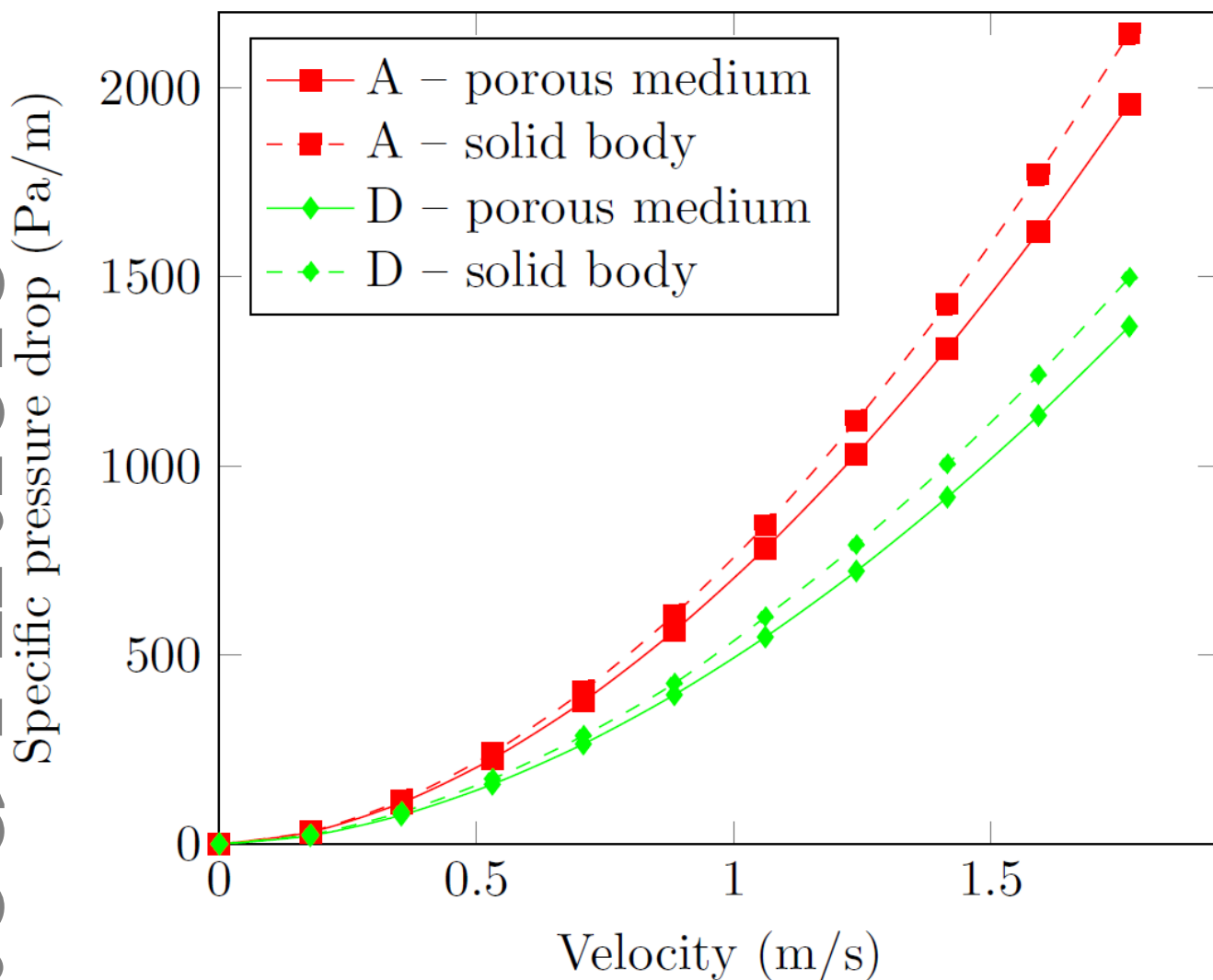


FIGURE 12 The variations of the specific pressure drop versus the velocity when the packed bed is set as a porous medium and as a solid body.

TABLE 1 The geometrical characteristics of catalysts

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diameter
A	25	25	–
B	25	25	1x13
C	25	25	7x5
D	25	–	7x5